



CLIMATE ACTION THROUGH IMPROVED COOKING ACCESS IN INDO-GANGETIC PLAINS

DOCUMENT PREPARED BY

SDG13 VENTURES PTE LTD

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The proposed grouped project activity seeks to reduce greenhouse gas emissions by disseminating fuel-efficient cookstoves in Uttar Pradesh, Uttarakhand & Bihar states of India.

Biomass has been considered as one of the primeval resources of energy and it is among the most favoured matter utilized by mankind since thousands of years. Fuel wood is one of the commonly used sources of domestic energy in the rural areas of many developing nations. In rural areas of Uttar Pradesh, Uttarakhand and Bihar, the primary source of fuel wood is cut down dry and diseased trees, fallen wood scraps or branches. Various studies¹² show that more than 70% of the rural population in the states targeted is dependent on fuel wood for cooking and other domestic purposes, which impacts the limited forest resources. The increase in demand for fuel wood degrades the biodiversity and the quality of Mother Nature. The studies reveal that felling of trees for fuelwood is far greater than the rate of vegetational growth.

'SDG13 Ventures Pte Ltd' (herein after called as "SDG") is project proponent ("PP") along with 'Offsetfarm Pte Ltd' (herein after called as "OPL") for this project. SDG & OPL are companies registered in the Republic of Singapore.

SDG through its representative in Host Country will coordinate with the users of ICS directly or through local partners or its representative to ensure that all requirements with respect to project activity, such as assisting with validation and registration, record keeping, monitoring and survey of households at a regular interval are met.

The individual households using ICS will sign an agreement with the project proponent acknowledging the inclusion of their stove into project activity.

¹ CONSUMPTION TRENDS OF FUEL WOOD AND THEIR IMPACT ON FOREST IN JHANSI DISTRICT, UTTAR PRADESH, INDIA

² How do People in Rural India Perceive Improved Stoves and Clean Fuel? Evidence from Uttar Pradesh and Uttarakhand



Figure 1: Snapshots of the stakeholders during the demonstration of the project ICS

The baseline scenario is the continued use of non-renewable wood fuel by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity.

The grouped project is expected to distribute 500,000 improved cook stoves in phased manner of distribution. It has planned to distribute 100,000 ICS/year for up to 5 years.

The estimated average annual GHG emission reduction from the project is 176,7067 tCO₂e and the total GHG emission reduction estimated for the whole crediting period from the project activity is 17,670,670 tCO₂e.

Audit Type	Period	Program	VVB Name	Number of years
Validation	(7 November, 2022 – 05 April 2023)	VCS	4K Earth Sciences Pvt. Ltd.	N/A for validation
Total	5 Months 7 November, 2022 – 05 April 2023			

1.2 Sectoral Scope and Project Type

Sectoral Scope: 03

Project Type: Type II – Energy Efficiency Improvement Project

Grouped Project: Yes

1.3 Project Eligibility

The project involves energy efficient cookstove distribution which falls under the category of efficiency improvements in thermal applications, therefore it is eligible under the scope of VCS Program.

1.4 Project Design

The project is a multiple project activity instance. Instances under this project consist of the distributed ICS with a thermal efficiency of at least 25% to household users cooking with non-renewable biomass in the baseline scenario.

☒ The project is a grouped project

▪ Eligibility Criteria

The project is a grouped project. Each new project will be confirmed against the eligibility criteria below:

No.	Eligibility criterion - Category	Eligibility criterion - Required condition	Supporting evidence for inclusion
1.	Defined geographical boundary	The new project instance boundary shall be confined to the state boundaries of Uttar Pradesh, Bihar & Uttarakhand.	Records of household location under the customer registration database. For this instance, records of sales (customer registration database) with clear delineation of project boundary i.e. the states of Uttar Pradesh, Uttarakhand and Bihar, is provided to the VVB for their reference.

No.	Eligibility criterion - Category	Eligibility criterion - Required condition	Supporting evidence for inclusion
2.	Conditional to avoid double counting	To prevent double counting of ICSs distributed under the project, ICS shall be uniquely marked, and new project instance implementers shall provide assurances/declarations that they are not included in any other CDM or Voluntary GHG offset schemes outside of this project	Unique marking on each ICS (such as a Logo/ Serial number/ Geo Coordinates/ Name of end user, address, batch number inter alia which identifies it as part of the new project instance together with the project database) Written attestation from the project proponent confirming that the ICSs distributed in the new project instance are not part of any other CDM or Voluntary GHG trading scheme.
3.	The specifications of technology/measures including the level and type of service, performance specifications including compliance with testing/certifications	All technologies implemented in the project must demonstrate a minimum thermal efficiency of 25%.	Independent Third-Party lab testing certificate demonstrating thermal efficiency greater than 25% using water boiling test or accepted national/international tests as allowed in CDM methodology AMSIIG. Product description and specifications For this instance, lab test efficiency certificate for the model distributed under this instance has clearly shown the efficiency above 25% has been submitted to VVB.
4.	The conditions that ensure that new project instances meet the requirements pertaining to the demonstration of additionality in VCS Standard.	Additionality shall be demonstrated as per the established methods in the applied version of methodology VMR0006.	There are no laws and regulations governing the use of improved cookstoves in India for households. The project is a voluntary effort by the project proponent. Though, the government of India promotes use of clean fuel e.g. LPG, however due to the initial capital cost and operating cost, the usage levels are low and most of the rural households still use traditional stoves for cooking. There is no specific concern made on improved cookstoves project from the above laws and regulations. Additionality for this instance has been demonstrated by using the investment barrier as the CDM

No.	Eligibility criterion - Category	Eligibility criterion - Required condition	Supporting evidence for inclusion
			Tool for demonstrating additionality. The future instances shall comply with the requirement of methodology for additionality demonstration.
5.	Conditions to provide an affirmation that funding from Annex I parties, if any, does not result in a diversion of official development assistance	Under this project, new project instance operators shall be required to provide an affirmation that funding from annex I parties, if any, does not result in a diversion of official development assistance	Individual new project instance operators shall sign ODA declarations for new project instances implemented by them to attest that no funding from annex 1 parties that leads to diversion of ODA has been used in the project. The statement shall specify whether or not annex 1 funding is used within the project Where funding is sought from annex 1 parties, the ODA declaration shall be supported by details regarding annex 1 funding that affirms that the funding doesn't divert ODA For this instance, no funding from Annex-1 parties is received and declaration to this effect has been furnished to VVB.
6.	The Group project-specific requirements stipulated by the project proponent including any conditions related to undertaking local stakeholder consultations and environmental impact analysis	The local stakeholder consultation will be conducted initially during the group PD level. Each new project instance/project will be implemented in similar social economic situations. The key stakeholders of the program both at group project activity and new project instance level are the same. No further actions needed at the new project instance level to satisfy the eligibility criteria.	The conditions to meet the requirements on undertaking the local stakeholder consultation have been proven in section further sections of this PD. The conditions to meet the requirements on undertaking the environmental impact assessment have been proven in the group PD in the section 2.3 of the VCS PD.
7.	Where applicable, the conditions related to sampling requirements for the group project	Monitoring of all new project instances will adhere to all requirements related to sampling for a group project activity in accordance with the sampling guidelines	Specification of the sampling methods applied and compliance with the sampling requirements are established at the group DD. For each new project instance, sampling will be undertaken as part of the group project activity

No.	Eligibility criterion - Category	Eligibility criterion - Required condition	Supporting evidence for inclusion
	in accordance with the "Standard for sampling and surveys for CDM project activities and programme of activities"	including all annexes and amendments till EB 86 Annex 04.	sampling plan, and in the new project instances shall describes how the group project activity sampling Plan is to be applied. For this instance, sampling plan is described in subsequent sections.
8.	Approval of new project instance by project proponent	The project proponent approves each project instance to be included into its registered group project activity.	Statement of project proponent giving approval for the new project instance to be included into its registered group project activity. Since, for this instance project proponent is the implementer, no further approval is required, however for the cases, where implementer other than PP, an approval on the same shall be given by PP in order to include the instance into this grouped project.
9.	VCU ownership	End users receiving ICSSs under the specific new project instance contractually cede their rights to claim and own emission reductions under the VCS to the project proponent of the group project.	A default distribution agreement for end users including the provision that emission reductions generated by the stove are owned by the project proponent will be provided for each new project instance. Waiver of carbon credits from the end users shall be submitted to VVB at the time of verification. For this instance, no distribution has been done so far and the start date is futuristic.
10.	Awareness and agreement of those operating a new project instance on group project subscription	Contractual provisions to ensure that those operating the new project instance are aware and have agreed that their activity is being subscribed to the new project instance.	Evidence for inclusion, in case new project instance operators are different from the project proponent: a declaration from project proponent/implementer, stating that they are aware and have agreed that their activity is being subscribed to the group project will be provided for each new project instance. Since, for this instance, project proponent is the only implementer, no further requirements related to awareness and agreement on

No.	Eligibility criterion - Category	Eligibility criterion - Required condition	Supporting evidence for inclusion
			their instance to be subscribed under this grouped project, however, for the cases, where implementer other than PP, an agreement on the awareness between PP and new instance implementer shall furnished.
11.	Efficiency of ICSs	Independent Third-Party lab testing certificate demonstrating thermal efficiency greater than 25% using water boiling test or accepted national/international tests as allowed in CDM methodology AMSIIG.	Model of ICS included in the group project shall comply the minimum efficiency 25% of applied methodology. The first instance has Greenway Jumbo model and the efficiency is defined ex-ante. For any new model of ICS, the efficiency shall be defined at the time of inclusion in group project.
12.	Baseline and applicability of methodology	The new instance shall adhere the baseline and applicability requirements of the VMR0006.	Baseline in conjunction with applicability assessment shall be part of each instance and adhere the requirements of VMR0006.
13.	The project proponent shall include in a singular project all project activity instances within ten kilometers of another instance of the same project activity and with the same project proponent (i.e., instances of the same project activity may not be spread across more than one project if they are within ten kilometers of each other).	If the distributions of ICSs are within the 10 Km range by the same project participants, they will be included in single instance.	The database with locations/address shall be shared with the VVB during the monitoring to comply the requirement.
14.	Where a capacity limit applies to a project activity included in the project, no project activity	There is no capacity limit as the project is developed as large-scale project. In case future instance has the capacity limits the requirements of VCS standard shall apply	The database with locations/address shall be shared with the VVB during the monitoring to comply the requirement.

No.	Eligibility criterion - Category	Eligibility criterion - Required condition	Supporting evidence for inclusion
	instance shall exceed such limit. Further, no single cluster of project activity instances shall exceed the capacity limit, determined as follows: 1) Each project activity instance that exceeds one percent of the capacity limit shall be identified. 2) Such instances shall be divided into clusters, whereby each cluster is comprised any system of such instances such that each instance is within one kilometer of at least one other instance in the cluster. Instances that are not within one kilometer of any other instance shall not be assigned to clusters. 3) None of the clusters shall exceed the capacity limit and no further project activity instances shall be added to the project that would cause any of the clusters to	as indicated in eligibility criteria.	

No.	Eligibility criterion - Category	Eligibility criterion - Required condition	Supporting evidence for inclusion
	exceed the capacity limit.		
15.	Start Date of the New Instance	All future instances shall have the start date after the date of first instance.	The evidence to support the start date of future instances shall be furnished to VVB.

1.5 Project Proponent

Project Proponent 1 & 2

Organization name	SDG13 Ventures Pte Ltd
Contact person	Mr Ankit Mathur
Title	Director
Address	100 Peck Seah Street #10-18 PS100 Singapore 079333
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Email	sdg13ventures@gmail.com

Organization name	Offsetfarm Pte Ltd
Contact person	Mr Saurabh Saraf
Title	Director
Address	160 Robinson Road, #14-04 Singapore Business Federation Center
Telephone	+65 6827 6828
Email	saurabh@offsetfarm.io

1.6 Other Entities Involved in the Project

N/A

1.7 Ownership

The project ownership is jointly with 'SDG13 Ventures Pte Ltd' & 'Offsetfarm Pte Ltd'. During the registration of ICS, the participating household will sign a declaration to transfer the ownership rights of the carbon assets generated from this project to the project proponent. The communication agreement between the entities shall be shared with VVB for validation and shall also be submitted to VERRA registry.

1.8 Project Start Date

01/10/2024 , the expected start date, it is assumed that the first ICS will be distributed by this date. However, this is an assumption based on the planning.

1.9 Project Crediting Period

01 Oct 2024 to 30 Sept 2034, ten years fixed crediting period.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale
Project

Large project	X
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Year	Estimated GHG emission reductions or removals (tCO ₂ e)
1.	657,835
2.	1306,533
3.	1,882,221
4.	2,393,088
5.	2,846,406
6.	2,590,799
7.	2,298,976
8.	1,723,288
9.	1,212,421
10.	7,59,103
Total estimated ERs	17,670,670
Total number of crediting years	10
Average annual ERs	1,767,067 (Rounded down)

1.11 Description of the Project Activity

Stoves disseminated under this grouped project activity will target rural and semi-urban households using traditional, unimproved cookstoves and open fires for cooking. The projects under the group project will distribute improved wood cookstove to households, a portable improved-efficiency stove manufactured in India. The ICS will have an average lifespan of 7 years and thermal efficiency of about 38%.

A typical project instance under project activity will replace conventional firewood stoves with higher efficiency ICS models by leveraging resources provided by the project activity in rural/semi-urban areas. ICSs are more efficient than conventional firewood stoves as they reduce heat loss and improve heat transfer and combustion efficiency. The ICS models are fuel efficient, resulting in a decrease in fuel use in comparison to conventional pre-project stoves while also reducing particulate matter and carbon emissions. Design considerations have also incorporated ergonomics and safety.

The project activity will continually assess biomass stove technology options with the goal of providing the high performing, affordable and locally appropriate technologies to the local environments when possible. As the project expands, several models of biomass stoves produced by ICS manufacturers may be included in the project activity. Inclusion of such stoves would be subject to compliance with requirements of the methodology and the eligibility criteria of the project. The PP is committed to investing in research and development for the improvement of the current stoves being proposed to be disseminated. Thus, during the life of the project activity, research and development may result in dissemination of more efficient ICS models, which shall be absorbed by this project, subject to methodological and eligibility criteria of the project. Upon inclusion into the project activity, all ICS will remain valid throughout the lifetime of the project period until the PP chooses to discontinue crediting of the improved stoves. Each project instances will provide a detailed description on the specific stove model/s implemented/envisaged to be implemented and the target customers for each of them. Greenway Jumbo stoves are the currently eligible ICS. The ICS cooking system is made of a fuel chamber for combusting the renewable biomass. The pot-rest is above the fuel chamber. The stove has one air inlet and weighs approximately 3782 grams.



Figure 2. Greenway Jumbo ICS

A record keeping system and unique identification of the ICS will be done by user's information collected through the Beneficiary Agreement and compiled in distribution records. This will include user name, identification number and address/location of the user's house, stove unique serial code and distribution date. The record keeping system will ensure that each ICS can be traced to project to avoid double counting.

1.12 Project Location

The ICSs in this PA are located at various locations across rural areas of Uttar Pradesh, Uttarakhand and Bihar. The geographical coordinates of Uttar Pradesh, Uttarakhand & Bihar of India are:

S.N.	States	Latitude	Longitude
1	Uttar Pradesh	23° 52'N to 31° 28'N	77° 3' to 84° 39'E
2	Uttarakhand	28° 44' & 31° 28' N	77° 35' & 81° 01' E
3	Bihar	24° 20' N to 27° 31'N	83° 19'E to 88° 17'E.

The map of Uttar Pradesh, Uttarakhand and Bihar showing the district coverage by the project is given below:

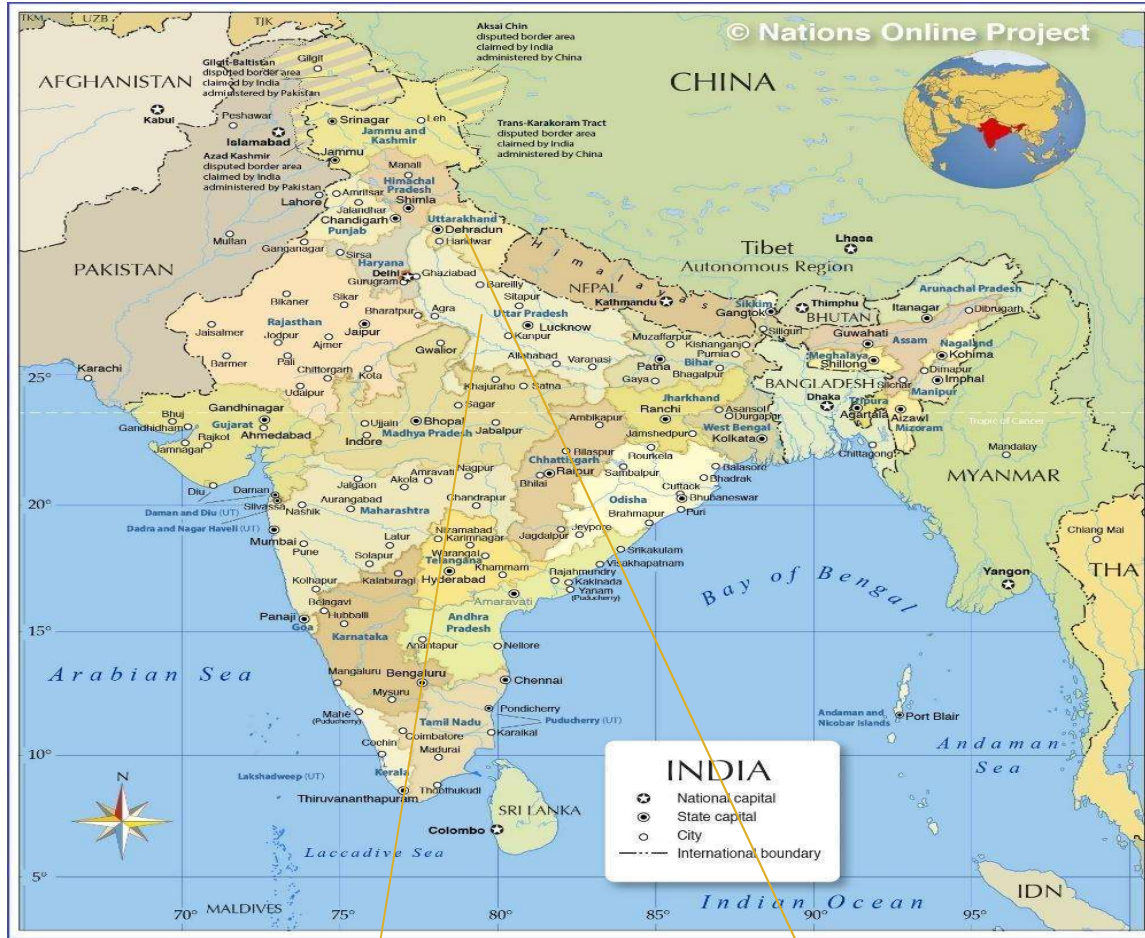


Figure 2. Map of India

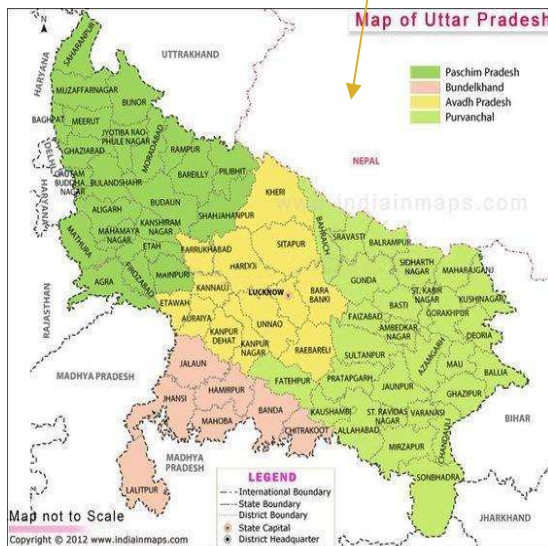


Figure 3. Map of Uttar Pradesh

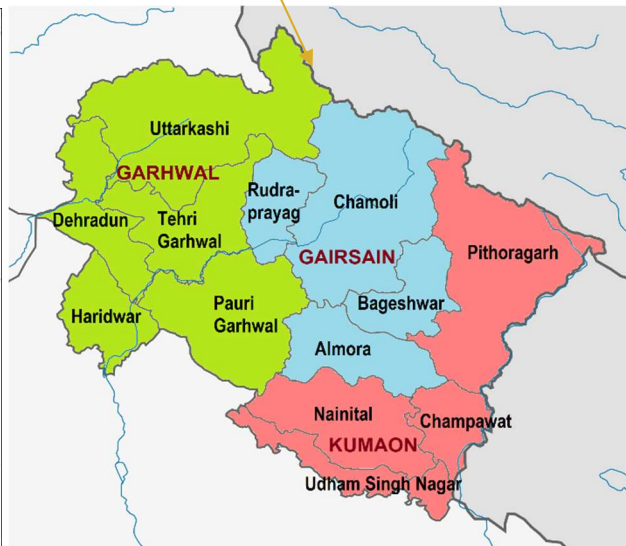


Figure 4. Map of Uttarakhand



Figure 5. Map of Bihar

1.13 Conditions Prior to Project Initiation

The conditions prior to project initiation are the continued use of non-renewable wood fuel (firewood) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity³. The project aims to reduce the environmental and social burdens caused by inefficient biomass cooking by introducing energy efficient wood cook stoves to low income households.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

There are no laws and regulations governing the use of improved cookstoves in India for households. The project is a voluntary effort by the project proponent. Though, the government of India promotes use of clean fuel e.g. LPG, however due to the initial capital cost and operating cost, the usage levels are low and most of the rural households still use traditional stoves for cooking.

There is no specific concern made on improved cookstoves project from the above laws and regulations.

1.15 Participation under Other GHG Programs

3

https://www.researchgate.net/publication/259959477_How_do_People_in_Rural_India_Perceive_Improved_Stoves_and_Clean_Fuel_Evidence_from_Uttar_Pradesh_and_Uttarakhand

<http://rchiips.org/nfhs/NFHS-4Reports/UttarPradesh.pdf>

<http://rchiips.org/nfhs/NFHS-5Reports/Uttarakhand.pdf>

<http://rchiips.org/nfhs/NFHS-5Reports/Bihar.pdf>

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project activity has not been registered or currently seeking registration under any other GHG programs. The project has not participated under any other GHG program.

1.15.2 Projects Rejected by Other GHG Programs

The Project Participant hereby declares that any other GHG program has not rejected the project activity; an undertaking for the same has been submitted by the Project Participant.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

This project activity is a voluntary initiative and it is not to meet any local laws or regulatory compliances. An undertaking has been submitted that PP shall not claim for GHG emission reduction credits for the given crediting period under any other emission-trading program or GHG binding limits.

1.16.2 Other Forms of Environmental Credit

The Project Participant declares that emission reductions generated from the project activity will not be double-counted (i.e. issuance of other forms of environmental credit like under CDM) for the particular crediting period, which is being claimed under VCS mechanism. PP has submitted an undertaking to the VVB that they shall not claim for GHG emission reduction credits for the given period under any other emission-trading program

Supply Chain (Scope 3) Emissions

N/A for the project.

1.17 Sustainable Development Contributions

The project activity includes distribution of improved cookstoves amongst the beneficiaries in the rural areas of Uttar Pradesh, Uttarakhand and Bihar, India, who were using the traditional cookstoves in the pre-project scenario, thereby, replacing the baseline inefficient cookstoves. Implementation of the project activity results in reduction of firewood consumption (through facilitation of fuel-efficient combustion), improving indoor air environment (by reducing emissions, smoke and soot resulting from inefficient firewood consumption) and employment generation for the local communities (in distribution, maintenance and monitoring of the ICS distributed). Such results and benefits contribute to the various sustainable development priorities of India.

The project's contribution towards sustainable development is established in line with the sustainable development indicators framed by India's stated sustainable development priorities, as set out in the SDG India Index Report 2021. The project activity contributes towards sustainable development of the region as well as of the nation, showing action on at least 7 different SDGs, as illustrated in figure 7.



Figure 6: Sustainable development contributions under different SDGs by the project activity

The significant sustainable development contributions envisaged by the project activity have been listed in table 1.

Table 1: A list of the relevant SDG targets and the related indicators of the project activity

SDG Targeted	Most relevant SDG Target	SDG Impact Indicator
1 No poverty	1.4 By 2030, ensure that all men and women, in particular the poor and the vulnerable, have equal rights to economic resources, as well as access to basic services, ownership and control over land, and other forms of property, inheritance, natural resources, appropriate new technology and financial services, including microfinance.	1.4.1 Proportion of population living in households with access to basic services Indicator: 1) Total number of ICS distributed under the project 2) % of users reporting money savings due to reduction in purchased fuel consumption
3 Good health and well being	3.9 By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination	3.9.1 Mortality rate attributed to household and ambient air pollution Indicator: % of users reporting reduction in smoke after shifting to cooking with the distributed ICS
5 Gender equality	5.4 Recognise and value unpaid care and domestic work through the provision of public services, infrastructure and social protection policies and the promotion of shared responsibility within the household and the family as nationally appropriate	5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age and location Indicator: % of users reporting time savings due to reduction in collecting fuelwood/ cooking time due to use of project ICS

7 Affordable and clean energy	7.1 By 2030, ensure universal access to affordable, reliable and modern energy services	7.1.2 Proportion of population with primary reliance on clean fuels and technology Indicator: % of users reporting operational project stove
8 Decent work and economic growth	8.3 Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises, including through access to financial services	8.3.1 Proportion of informal employment in total employment, by sector and sex Indicator: Number of employment created by project
10 Reduce inequality within and among countries	10.1 By 2030, progressively achieve and sustain income growth of the bottom 40 percent of the population at a rate higher than the national average	10.1.1 Growth rates of household expenditure or income per capita among the bottom 40 percent of the population Indicator: Financial and time savings reported by end users of the project
10 Reduce inequality within and among countries	10.2 By 2030, empower and promote the social, economic and political inclusion of all, irrespective of age, sex, disability, race, ethnicity, origin, religion or economic or other status	10.2.1 Proportion of people living below 50 percent of median income, by sex, age and persons with disabilities Indicator: Number of female local partners employed in the implementation of the project
13 Climate action	13.2 Integrate climate change measures into national policies, strategies and planning	13.2.2 Total greenhouse gas emissions per year Indicator: Amount of net CO ₂ e emissions reduced by the project per year

1.18 Additional Information Relevant to the Project

- [Leakage Management](#)

Not applicable as the project adopts a net gross adjustment factor of 95% to account for leakage.

- [Commercially Sensitive Information](#)

No commercially sensitive information has been excluded from the public version of the project description.

- [Further Information](#)

Not Applicable

2 SAFEGUARDS

2.1 No Net Harm

No potential negative environmental or socio-economic impacts have been identified for the project. The project involves the distribution and installation of Greenway certified improved cook stoves to households in the states of Uttar Pradesh, Uttarakhand and Bihar, in India, which currently use mostly traditional open fire for cooking. The activities under the proposed project promote improved cook stoves that result in reduced fuel consumption and emissions due to cooking and heating water in homes. The ICS used in this project have characteristics that improve the efficiency of combustion and thermal transfer to the pot compared with three-stone fires or traditional pot support. Furthermore, traditional cooking methods involve open fires that result in the emissions of local pollutants such as carbon monoxide and particulate matter in often poorly ventilated rooms, which lead to respiratory problems. In addition, open fires are frequent causes of burns and other injuries. Switching from fireplaces to ICS reduces the incidence of such injuries and health problems. Therefore, the Greenway Jumbo Stove installed under this project presents positive environmental impacts wherever they are applied, and no negative environmental impacts have been identified.

2.2 Local Stakeholder Consultation

The local stakeholder consultations (LSC) for the grouped project activity were held, both physically and virtually, with the objective of gathering comments and concerns of the stakeholders on the proposed project activity. The stakeholder consultation was conducted at the grouped project activity level as the technology and its impacts will be similar for all the project activities implemented in this grouped project activity. The details of the LSC meetings held are as follows:

Sl. No.	Location	Mode- Physical/ virtual	Date	Time
1	Google meet	Virtual	25 November, 2022	11 am to 12 pm
2	Boond Bachat Sangh, Kulgaon, Kanpur district, Uttar Pradesh	Physical	25 November, 2022	12:15 pm to 1:30 pm
3	Anganwadi Center of Lalpur Village, Kanpur district, Uttar Pradesh	Physical	25 November, 2022	2 pm to 3:30 pm

The objective was to involve as many stakeholders as possible, through both the physical and virtual consultations, and they were informed through official emails for the virtual LSC and press release in the state daily “Aj”, whose circulation spans the states of Uttar Pradesh, Uttarakhand and Bihar, on 29 October, 2022 (figure 7) for both the physical LSCs. All interested stakeholders from the following groups: traditional stove users, local and international NGO representatives, government organisations, social impact companies, carbon market experts, financing companies, and field assistants working in the project, attended the meetings (figures 8 and 9). Over 52 participants in total had attended the virtual and physical LSC meetings. The physical meetings were attended by over 41 participants and the virtual meeting was attended by 11 participants.

The consultation meetings started with a brief explanation of the agenda of the public consultation to the participants and they were encouraged to share their opinions on the upcoming cookstove program and the social, economic and environmental impacts created through it. An informative presentation on traditional fuel types that are being used for daily cooking in households and the role that it plays in global warming and impacts on human health was also conducted. The phenomenon of climate change and the impact of household fuelwood consumption in the increased levels of GHGs was also discussed. The LSC attendees were presented with the Greenway Jumbo stove and were told about its advantages- the positive impacts the activity was expected to create in the project location, including the cost-effectiveness, health benefits to the beneficiaries and amount of time that would be saved through reduced need for collection of fuelwood. In addition, the various SDG impacts that would be achieved through the project and the product subsidization to be provided to the beneficiaries was also discussed.

The participants asked questions about how to use the stoves, its benefits, the cost of it, and discussed the potential application or deployment areas. They remarked that it would be a useful device to be used in the rural and semi-urban areas where women are using wood based cookstoves. It was also discussed how the current SHG group members can be employed in the distribution of the devices and that would create employment opportunities for everyone involved.

आज

महानगर

कागज़: 29 अक्टूबर 2022

आईआईटी में केवि व शॉपिंग सेंटर के इर्द-गिर्द दिखा तेंदुआ

शुक्रवार को केवि में छुट्टी, लेकिन शानिवार को खुलेगा: प्रिंसिपल



(आज कागज़: 29 अक्टूबर) आईआईटी के शॉपिंग सेंटर के इर्द-गिर्द दिखा तेंदुआ (Asian Monoceros) की तस्वीरें सामने आई हैं। शॉपिंग सेंटर के इर्द-गिर्द दिखा तेंदुआ (Asian Monoceros) की तस्वीरें सामने आई हैं। शॉपिंग सेंटर के इर्द-गिर्द दिखा तेंदुआ (Asian Monoceros) की तस्वीरें सामने आई हैं।

आईआईटी के प्रिंसिपल ने कहा कि शनिवार को शॉपिंग सेंटर खुलेगा। उन्होंने कहा कि शनिवार को शॉपिंग सेंटर खुलेगा। उन्होंने कहा कि शनिवार को शॉपिंग सेंटर खुलेगा।

रफ्तार का कहर, हादसों में तीन की दर्दनाक मौत

कागज़: 29 अक्टूबर 2022



रफ्तार का कहर, हादसों में तीन की दर्दनाक मौत

लाल चकत्ते व ब्लीडिंग के साथ डेंगू हुआ जानलेवा

हैलट एंव चर्चला ओपीकी में बुखार के रोगियों की भरमार, गंभीर रोगी किये गये भर्ती



लाल चकत्ते व ब्लीडिंग के साथ डेंगू हुआ जानलेवा

हैलट एंव चर्चला ओपीकी में बुखार के रोगियों की भरमार, गंभीर रोगी किये गये भर्ती

12 नये डेंगू रोगी मिले जिले में अब 134 पॉजिटिव केस

कागज़: 29 अक्टूबर 2022

12 नये डेंगू रोगी मिले जिले में अब 134 पॉजिटिव केस

चौकी प्रभारी पर समझौते का दावा बनाने का आरोप

कागज़: 29 अक्टूबर 2022

चौकी प्रभारी पर समझौते का दावा बनाने का आरोप

माफूसी बात पर नाराजों ने बुक को पीटा

कागज़: 29 अक्टूबर 2022

माफूसी बात पर नाराजों ने बुक को पीटा

सेक्टरम तेवत तक पहुंचा मेट्रो स्टेशन का निर्माण

टॉप हाउस प्रभासी से बन रहा भूमिगत स्टेशन, रुकड़ का कांस्ट्रिक्ट



सेक्टरम तेवत तक पहुंचा मेट्रो स्टेशन का निर्माण

टॉप हाउस प्रभासी से बन रहा भूमिगत स्टेशन, रुकड़ का कांस्ट्रिक्ट

साइबर ठगों ने बुक को तगवा हजारां का चूना

कागज़: 29 अक्टूबर 2022

साइबर ठगों ने बुक को तगवा हजारां का चूना

जहर से मौत की आशंका पर बिसरा किया सुरक्षित

कागज़: 29 अक्टूबर 2022

जहर से मौत की आशंका पर बिसरा किया सुरक्षित

हा की हल्ला में भांजे ने मामा पर लगाया था आरोप

कागज़: 29 अक्टूबर 2022

हा की हल्ला में भांजे ने मामा पर लगाया था आरोप

नाना और सात्या स्ट्रीटमल शॉप्ट के बाद बनेगा स्टेशन का हिस्सा

कागज़: 29 अक्टूबर 2022

नाना और सात्या स्ट्रीटमल शॉप्ट के बाद बनेगा स्टेशन का हिस्सा

कागज़: 29 अक्टूबर 2022

कागज़: 29 अक्टूबर 2022

कागज़: 29 अक्टूबर 2022

स्थानीय हितधारक परामर्श बैठक

Project: "Climate action through Improved cooking access in Indo-Gangetic plains"

कार्य सूची : प्रस्तावित परियोजना के परियोजना डिजाइन, प्रभावों और लाभों पर चर्चा

SDG13 Ventures और OffsetFarm की पहल जिसका उद्देश्य महिलाओं के साथ काम करना तथा स्वच्छ चूल्हों के वितरण को आगे बढ़ाना है। SDG13 Ventures परियोजना को Verra VCS रजिस्ट्री में पंजीकृत करेंगे।

दिनांक: 25 नवंबर 2022 समय: 12:30 से

SDG13 OffsetFarm शारदाक भवार्थी
अंगिक भारतीय

Figure 7: Advertisement placed in a state daily "Aj" on 29/10/22 for the physical Local Stakeholder Consultation



Figure 8: A snapshot of the physical LSC held at Boond Bachat Sangh, Kulgaon, Kanpur district, Uttar Pradesh (25/11/22)



Figure 9: A snapshot of the physical LSC held at Anganwadi Center of Lalpur village, Kanpur district, Uttar Pradesh (25/11/22)

The grievance redressal process and the steps that would be taken to look into and rectify the issues being faced by the beneficiaries of the project ICS was thoroughly explained to the stakeholders involved. The nature of customer relationships that would be maintained was also explained to the stakeholders. All the attendees were provided with the details of the project implementors if they wished to raise any concerns or questions even after the stakeholder meetings. No negative comments were raised by the stakeholders during the meetings and due to the associated benefits, the stakeholders have appreciated the proposed project activity.

Ongoing Feedback Mechanism:

The provision of continuous feedback will be ensured with the project monitoring surveys as part of monitoring of GHG related parameters, also the grievance register shall be placed at the local offices of the SDG 13 Ventures Pte. Ltd.

2.3 Environmental Impact

No negative environmental impacts have been identified from the project and environmental impact assessment (EIA) is not required for the project.

There are no major impacts on the environment due to the installation of the ICS. The local ecology is not likely to get impacted by this type of programme activity. The group activity included under this group project helps in reducing the consumption of the firewood thus reducing the pressure on forest, reduces indoor air pollution and benefits with the use of dung cakes as fuel farmyard manure thus having positive environmental impacts.

2.4 Public Comments

This project was open for public comment from 7 November, 2022 to 7 December, 2022. No comments were received during the public comment period.

2.5 AFOLU-Specific Safeguards

This section is not applicable as the project is a non-AFOLU project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Following approved baseline & monitoring methodology is applied;

Methodology: VMR0006- Methodology for Installation of High Efficiency Firewood Cookstoves

Version: 1.1

Sectoral Scope: 03

Tools:

“CDM TOOL30: Calculation of the fraction of non-renewable biomass”; Version-03.0 , EB 108, Annex 11

CDM Tool 1: Methodological tool Tool for the demonstration and assessment of additionality, Version 07

3.2 Application of Methodology

The methodology measures below constitute the justification for the choice of the selected methodology by showing that project instances under the project activity meets each applicability condition of the methodology.

S. No	VCS Methodology requirement	Project justification
1	The project stove is a single pot or multi pot portable or an in-situ cookstove using only woody biomass; Additional requirement to demonstrate that the biomass used is solely renewable biomass in case project activities replacing baseline stoves using fossil fuel	The proposed instances will distribute high efficiency single-pot improved cook stove for thermal application, which will replace inefficient traditional cookstove leading to saving of non-renewable biomass. The ICS to be distributed under the project activity is Greenway Jumbo stoves/greenway stoves. All ICS in this project will replace existing traditional wood stoves; therefore, no greenfield installations are included. Further, the project activity will utilize only non-renewable biomass under project scenario.

		This criterion will be checked from data recorded on the baseline stove used prior to ICS installation.
2	Project activities shall be implemented in domestic premises or in community-based kitchens	The project activity will replace traditional wood-based stove in individual households in rural/semi urban areas.
3	The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer's specifications and shall exclusively use woody biomass and can be single pot or multi-pot; in case of project stove replacing fossil fuel baseline stove, it shall exclusively use renewable biomass	The Greenway or Greenway Jumbo single pot portable improved cooking stoves planned to be distributed have a specified efficiency of 38% as per water boiling test in compliance with Tier II and above as per ISO as tested and certified by third party.
4	Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics	Proof that biomass has been used in the project region, i.e., in the host country India since 31 December 1989 is available through widespread documentation. Report by National sample survey organization (NSSO), department of statistics, Government of India reveals that 79% of the rural households in India were dependent of firewood and chips in the year 1987-88⁴. The same article reported 78% of rural households to be dependent on firewood and chips in the year 1993-94. National sample survey organization, Government of India reveals that 75% of the rural households in rural India were primarily dependent of fuelwood and chips⁵. Fuel wood is the main source of energy in rural areas. Wood energy is mainly used as fuel for cooking at household. The above clearly shows that the rural areas in the host country are dependent on biomass since and before 1989 with not much change in the trend which clearly depicts the dependence of the people of the host country on woody biomass as fuel for meeting thermal energy requirement for cooking at households.

⁴ <https://www.mospi.gov.in/publication/energy-statistics-india-2022>

⁵ http://mospi.nic.in/sites/default/files/publication_reports/511_final.pdf

5	For the specific case of biomass residues processed as a fuel (e.g. briquettes, wood chips), it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g. shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded	The ICS is introduced as energy efficiency measure to replace baseline stoves and reduce the use of non-renewable biomass for combustion. The consumption of the fuel used in the project activity will be monitored. If briquettes utilization in project case the energy consumption for manufacturing of briquettes and transportation of briquette will be monitored to calculate project emissions
6	The project description shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo).	A record keeping system and unique identification of the project device will be used. This will include user name, user identification number and address/location of the user's house and distribution date. The record keeping system will ensure that each ICS can be traced to project instances to avoid double counting.
7	The project description shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	The project proponent will sign an end user agreement with individual household/user to ensure only PP can claim the emission reduction. A default distribution agreement for end users including the provision that emission reductions generated by the stove are owned by the PP will be provided for each user.

Tool 30 - Calculation of the fraction of non-renewable biomass

Applicability criterion	How the project complies
This tool may be used by: (a) DNAs to submit region- or country-specific default f_{NRB} values, following the procedures for development, revision, clarification and update of standardized baselines (SB procedures);	Option (b) applied to calculate the project specific f_{NRB} value as described under section 4.1 of the PD.

or	
(b) project participants to calculate project- or PoA-specific fNRB values.	

CDM Tool 1: Methodological tool Tool for the demonstration and assessment of additionality, Version 07:

The compliance of the tool is demonstrated in the section 3.5 of VCS PD.

3.3 Project Boundary

The boundary of the group project is within the rural areas under various districts of state of Uttar Pradesh, Uttarakhand and Bihar, maps are presented below:

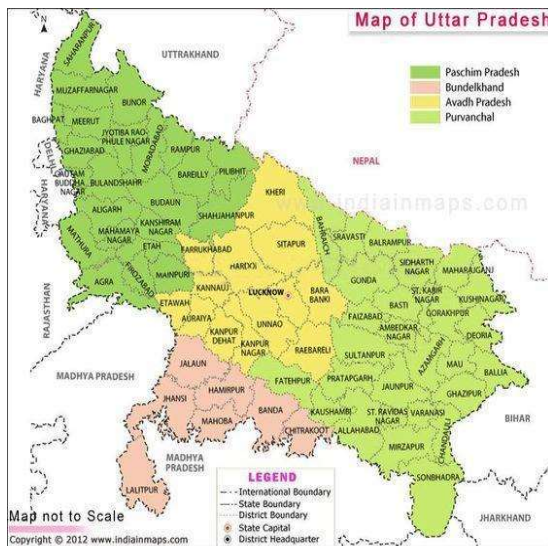


Figure 10: Map of Uttar Pradesh

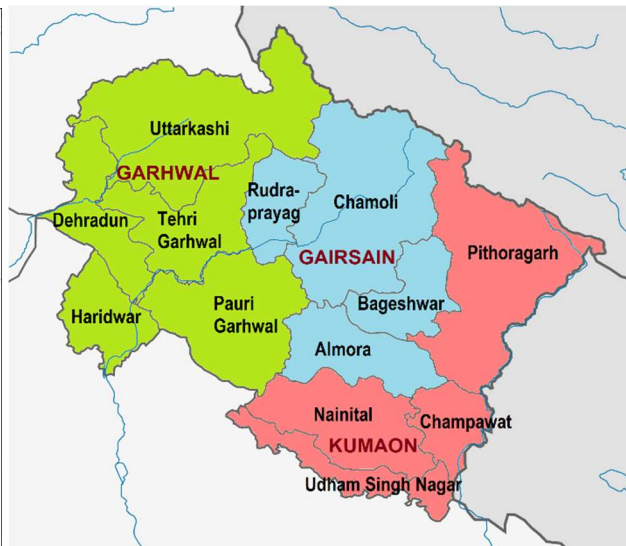


Figure 11: Map of Uttarakhand



Figure 12: Map of Bihar

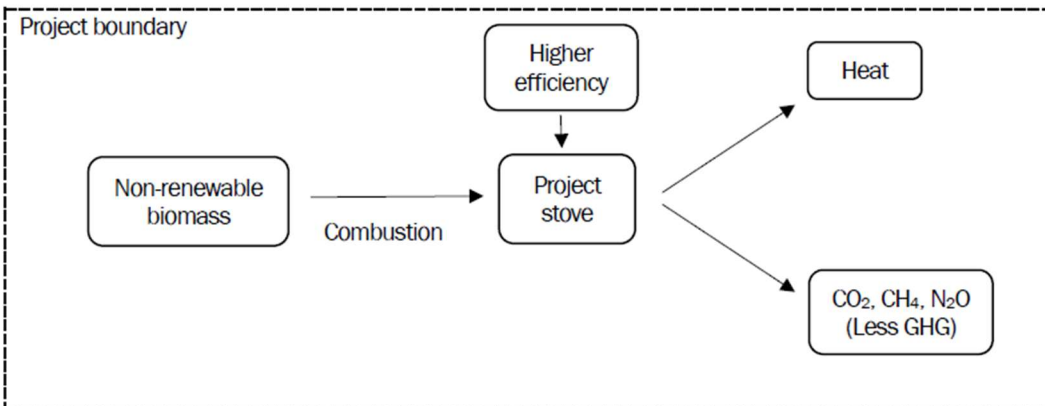
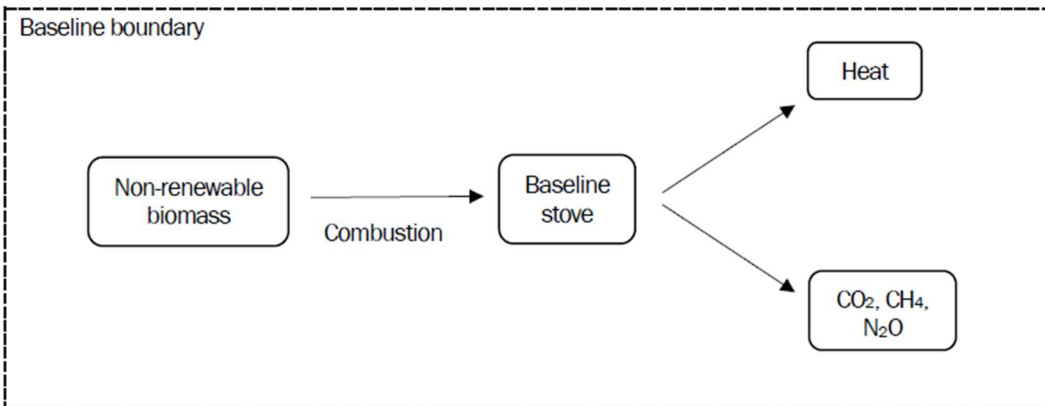


Table 1: GHG Sources Included In or Excluded From the Project Boundary

Source		Gas	Included ?	Justification/Explanation
Base line	Emission from use of non-renewable biomass/Fossil fuel	CO2	Yes	Major source
		CH4	Yes	Major source
		N2O	Yes	Major source
		Other	No	No other source identified
	Production & Transport of Fuel	CO2	No	Baseline and project fuels are same thus no transportation and production triggering from project activity.
		CH4	No	Baseline and project fuels are same thus no transportation and production triggering from project activity.
		N2O	No	Baseline and project fuels are same thus no transportation and production triggering from project activity.
		Other	No	Baseline and project fuels are same thus no transportation and production triggering from project activity.
Project	Emission from use of non-renewable biomass	CO2	Yes	Major source
		CH4	Yes	Major source
		N2O	Yes	Major source
		Other	No	No other source identified
	Production & Transport of Fuel	CO2	No	Baseline and project fuels are same thus no transportation and production triggering from project activity.
		CH4	No	Baseline and project fuels are same thus no transportation and production triggering from project activity.
		N2O	No	Baseline and project fuels are same thus no transportation and production triggering from project activity.
		Other	No	Baseline and project fuels are

				same thus no transportation and production triggering from project activity.
--	--	--	--	--

3.4 Baseline Scenario

The conditions prior to project initiation are the continued use of non-renewable wood fuel (firewood) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity⁶. The project aims to reduce the environmental and social burdens caused by inefficient biomass cooking by introducing energy efficient wood cook stoves to low-income households.

Table 2 illustrates that firewood consumption is the baseline fuel used by the rural and semi-urban populations in the states of Uttar Pradesh, Uttarakhand and Bihar, while chullah or traditional stoves is the baseline cooking technology, as sourced from the National Family Health Survey⁷ official reports.

Table 2: Types of cooking fuel and stoves in use in the project area

Variable	Uttar Pradesh ⁸	Uttarakhand ⁹	Bihar ¹⁰
Cooking fuel			
% using firewood	48.3	54.9	28.4
% using dung cakes	29.6	1.6	20.0
% using kerosene	0.1	0.0	0.0
% using LPG	0.1	42.2	29.8
% using	1.1	0.0	7.2

⁶

https://www.researchgate.net/publication/259959477_How_do_People_in_Rural_India_Perceive_Improved_Stoves_and_Clean_Fuel_Evidence_from_Uttar_Pradesh_and_Uttarakhand

⁷ http://rchiips.org/nfhs/NFHS-5_State_Report.shtml

⁸ <http://rchiips.org/nfhs/NFHS-4Reports/UttarPradesh.pdf>

⁹ <http://rchiips.org/nfhs/NFHS-5Reports/Uttarakhand.pdf>

¹⁰ <http://rchiips.org/nfhs/NFHS-5Reports/Bihar.pdf>

straw/shrub/grass			
% using clean fuel for cooking	16.4	42.9	30.3
% using solid fuel for cooking	83.3	56.8	69.5
Type of stove			
% households using stove	0.4	0.1	0.7
% households using chullah	99.1	99.5	98.9
% using open fire	0.4	0.4	0.4

3.5 Additionality

Activity Method

Step 1: Regulatory Surplus

There is no mandatory legal requirement for installation of cookstoves in households/institutions of Uttar Pradesh, Uttarakhand and Bihar, India. The PPs confirm that the grouped VCS project, and all actions taken as part of it, are voluntary and coordinated. A review of the national energy policies of the Uttar Pradesh, Uttarakhand and Bihar, India shows that there are no mandatory laws, policies or requirements mandating the use of ICS. The government promotes the use of improved ICS, however it's not mandatory. The energy policies are governed at the national level and the 'Energy Conservation Act 2001' and its 2022 amendment¹¹ governs the energy sector. The referred policies are the only policies which governs the energy sector in India.

The cited act does not enforce the PP to implement the ICS.

Step 2: Positive List

Criteria:

The applicability conditions of this methodology represent the positive list. The project must demonstrate that it meets all of the applicability conditions of the methodology as well as the below condition. In so doing, the project is deemed as complying with the positive list.

1. Where the project activity installs or distributes stoves at zero cost to the end-user and has no other source of revenue other than the sale of GHG credits, the project activity shall be deemed additional.

Justification: The PP is charging a minimal amount of 500 INR as administrative cost from the households. The cost is charged in order to avoid misuse of ICS, which is potentially possible if the ICS are distributed at zero cost.

Furthermore, through a simple cost analysis as shown in sub step 2b(i) under step 3: Project Method given below, it has been shown that the total cost of each ICS is valued at INR 2999 and it is being made available to the end users at ~80% subsidized rate for covering the administrative costs and preventing misuse of the distributed ICS.

2. Project activities that are implemented as part of government schemes or are supported by multilateral funds cannot be considered additional even if the stoves are distributed free of cost or at a highly subsidized rate and hence are not eligible to use this methodology.

Justification: The project activity is voluntary activity and is not part of any government scheme, the PP is charging a minimal amount of around 500 INR as administrative cost from the households. The cost is charged just to avoid misuse of ICS, which is potentially possible if the ICS are distributed at zero cost. People can just collect and do not use it. Notwithstanding, PP has used the project method too to demonstrate the additionality of project below.

¹¹ <https://egazette.nic.in/WriteReadData/2022/241246.pdf>

Step 3: Project Method

The “Tool for demonstration and assessment of additionality” Version-7.0.0 EB 70 Annex-08 is

applied to demonstrate additionality

The Tool provides for a step-wise approach to demonstrate and assess additionality.

These Steps

include:

Step 0- Demonstration whether the proposed project activity is the first-of-its-kind

Step 1- Identification of alternatives to the project activity consistent with current laws and

regulations

Step 2. Investment analysis

Step 3: Barrier analysis

Step 4: Common practice analysis

Step 0- Demonstration whether the proposed project activity is the first-of-its-kind

The proposed project activity is not first of its kind in host country.

Step 1: Identification of alternatives to the project activity consistent with current laws and

regulations

Sub-step 1a: Define alternatives to the project activity:

The project activity is to distribute improved cookstove to individual households to traditional stoves, the alternatives to the project activity are

a) *The proposed project activity undertaken without being registered as a VCS project activity;*

b) Continuation of the current situation.

Outcome of Step 1a: Identified realistic and credible alternative scenario(s) to the project activity

The alternative identified are

(a) The proposed project activity undertaken without being registered as a VCS project activity;

(c) Continuation of the current situation.

Sub-step 1b: Consistency with mandatory laws and regulations:

There are no policies in host country to regulate fuel usage for cooking at household level and both the alternative identified above are complying in host country rule and regulations.

Outcome of Step 1b: Identified realistic and credible alternative scenario(s) to the project activity that are in compliance with mandatory legislation and regulations taking into account the enforcement in the region or country and EB decisions on national and/or sectoral policies and regulations

Both the alternative identified above are complying with host country regulation.

Step 2: Investment analysis

To conduct the investment analysis, use the following Sub-steps:

Sub-step 2a: Determine appropriate analysis method

As the project proponent will sale ICS at a lower price than purchase cost of ICS and there are no other economic benefit than sale of VCUs, hence a simple cost analysis is chosen by PP.

Sub-step 2b: Option I. Apply simple cost analysis

The proposed project intends to distribute the ICS at ~80% subsidy to users i.e., households; hence financial return from the programme will be less than cost incurred, further the project proponent will hire a team for supply chain and maintenance and repair. As each ICS has certain costs associated with the production/purchase, distribution, monitoring and verification etc., the programme will not occur in absence of the carbon revenue. The action is not financially viable without the support of revenues from the sale of VCUs. As the end-user does not benefit from a direct financial return and procuring ICSs requires capital, which is a barrier to rural/semi-urban consumers due to difficulties in accessing capital, a wide dissemination of ICSs in the Host Country is unlikely. The actions under the project will alleviate these barriers by promoting distribution of ICSs to end-users at an affordable price. Hence, the project faces investment barrier and is considered to be additional. The simple cost analysis is substantiated below:

Actual Cost of Sale of ICS = 2999 INR/ICS Based on the actual sale to other Customer

Subsidized Sale to Households = 500 INR/ICS As per the framework of group VCS project

Loss = 2499 INR/ICS Without Carbon Revenue.

Alternate: Benchmark & IRR Analysis

The project has only one cash inflow and outflow therefore the simple cost analysis and IRR will have the similar results in terms of percentage loss but will additional comparison with benchmark and sensitivity.

Therefore, to further strengthen the additionality argument, an investment analysis in terms of IRR, sensitivity analysis has also been computed and included in the ER/IRR sheet submitted. The same is also presented below and also included in ER/IRR spreadsheet:

Assumptions	Currency	Source	Remark where applicable
Cost of ICS available at the time of investment decision	-INR 2,999.00	Invoice issued by Greenway to other Customer at the time of investment decision.	
Cost of ICS charged to user	INR 500.00	Investment decision	
Total ICS to be sold (Tragetted)	5,00,000.00	Investment decision	
Total Investment (Cash Outflow)	-INR 1,49,95,00,000.00	calculated	
Cash Inflow	INR 25,00,00,000.00	calculated	
IRR	-INR 0.83	calculated	
Loss per ICS	-INR 2,499.00		
Model of Finance	100% Equity		
Benchmark	Default Value of RoE as per the Tool 27, version 11 (available at the time of investment decision and valid at the time of RfR)	Long Term Inflation (Conservatively considered the lower value)	
For Singapore (PPs are the registered firm in Singapore. Moreover, the Singapore inflation forecast is lower than India therefore it is conservative too)	8.70%	1.48%	https://www.statista.com/statistics/379423/inflation-rate-in-singapore/

<u>Sensitivity Analysis</u>			
		Variation in project cost	
		Variation in subsidized cost	
Benchmark	10.18%		
IRR	-83.33%		
SUMMARY OF SENSITIVITY ANALYSIS:			
Variation in project cost	-10%	0%	10%
IRR	-83.33%	-89.58%	-83.33%
Variation in Subsidized Cost	-84.99%	0%	-81.66%

It is evident from the above analysis that the project is not financially attractive without the carbon revenue.

Step 3: Barrier analysis:

Not chosen by PP.

Step 4: Common Practice Analysis:

As per information available in public domain¹², the penetration level of ICS is very low in rural areas of Uttar Pradesh, Uttarakhand and Bihar, and majority of the population is still using the inefficient firewood cooking practices and project activity is not a common practice.

As each ICS has costs associated with its purchase and distribution and awareness generation, this programme will not occur in absence of the VCS revenue. The action is not financially viable without the support of revenues from the sale of VCU. The end-users for the ICS are currently using crude devices, which indicates and exacerbates the cycle of poverty/suppressed incomes.

The potential users for ICS are typically lower income with the problem (lack of ICS) seen more as a woman's problem thereby relegating it lower in the priority order for household expenditures/investments. The actions under the project will alleviate these barriers by promoting distribution of ICSs to end-users at upwards of an ~80% subsidy, thereby making their ownership and subsequent impact possible.

¹² [How do People in Rural India Perceive Improved Stoves and Clean Fuel? Evidence from Uttar Pradesh and Uttarakhand](#)

<http://rchiips.org/nfhs/NFHS-4Reports/UttarPradesh.pdf>

<http://rchiips.org/nfhs/NFHS-5Reports/Uttarakhand.pdf>

<http://rchiips.org/nfhs/NFHS-5Reports/Bihar.pdf>

3.6 Methodology Deviations

The project activity has not taken any deviation in applied approved methodology VMR0006, version 1.1.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The improved cookstove is introduced as energy efficiency measure in the project, therefore equations 1 and 2 of the methodology will be applied to calculate the net GHG emission reductions.

$$ER_y = \sum_i \sum_j ER_{y,i,j} \quad \text{Equation (2)}$$

Where:

- i = Indices for the situation where more than one type/model of improved cookstove is introduced to replace three-stone fire
- j = Indices for the situation where there is more than one batch of improved cookstove of type i
- ER_y = Emission reductions during year y in t CO₂e
- $ER_{y,i,j}$ = Emission reductions by improved cookstove of type i and batch j during year y in t CO₂e

$$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{wood\ fuel} \times f_{NRB,y} \times (EF_{wf,CO_2} + EF_{wf,non\ CO_2}) \times N_{y,i,j} \times 0.95 \quad \text{Equation (2)}$$

Where:

- $B_{y,savings,i,j}$ = Quantity of woody biomass that is saved in tonnes per improved cookstove of type i and batch j during year y
- $f_{NRB,y}$ = Fraction of woody biomass that can be established as non-renewable biomass (f_{NRB})

$NCV_{wood\ fuel}$	=	Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/tonne) ¹³
EF_{wf,CO_2}	=	CO ₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 112 tCO ₂ /TJ) ¹⁴
$EF_{wf,non\ CO_2}$	=	Non-CO ₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 26.23 tCO ₂ /TJ) ¹⁵
$N_{y,i,j}$	=	Number of improved cookstoves of type i and batch j operating during year y
0.95	=	Discount factor to account for leakage

To calculate $B_{y,savings}$, we use **equation 4** of the applied methodology¹⁶

$$B_{y,savings,i,j} = B_{y=1,new,i,survey} \times \left(\frac{\eta_{new,i,j}}{\eta_{old}} - 1 \right)$$

Where:

$B_{y=1,new,i,j,survey}$: Annual quantity of woody biomass used by improved cookstoves in tonnes per device of type i and batch j, determined in the first year of

the implementation of the project through a sample survey η_{old} . A default value of 0.10 may be optionally used if the replaced system is a three stone fire, or a conventional system with no improved combustion air supply or flue gas ventilation system, i.e., without a grate or a chimney.

$\eta_{new,i,j}$ Efficiency of the system being deployed as part of the project activity (fraction), as determined using the Water Boiling Test (WBT).

As some of the project households continue to use baseline cookstoves along with improved cookstoves, $B_{y,new\ survey}$ has been adjusted ex-post based on the percentage of project households found to continue such practice according to the Equation 6 of applied methodology as follows:

¹³ 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 1 Introduction (No change in 2019 refinements)

¹⁴ 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion (No change in 2019 refinements)

¹⁵ 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion (No change in 2019 refinements)

¹⁶ Equation 3 of methodology VMR 0006

For ex-ante estimates:

ER Estimation for Project Activity Instance:

For ex-ante calculation purpose, the assumption below is applied for first project activity instance.

- The group project under the first instance is planning to install ICS 100,000 ICS annually with total 500,000 distributions in five years.
- The life span of ICS is 7 years;
- $B_{y=1, \text{new}, i, \text{survey}}$ is assumed as 1.3 tonnes/device/year, which is a sole assumption based on the professional judgement of PP based on the 35% energy saving in project scenario. The actual quantity shall be monitored ex-post during the first verification of project

Calculation of f_{NRB}

Determination of the Share of Non-Renewable Biomass

According to methodological tool "Calculation of the fraction of non-renewable biomass" version 03.0, , equation 1, shall be used to calculate f_{NRB} :

$$f_{NRB} = \frac{NRB}{NRB + RB}$$

Where:

$f_{NRB,y}$: fraction of non-renewable biomass (%)

NRB : non renewable biomass (tons)

RB : Renewable biomass (tons)

For f_{NRB} calculation details the separate excel sheet is prepared in accordance with the CDM Tool 30 and shall be submitted to VVB for validation.

Particulars	Symbol	Unit	Values	Notes/Source
Average household wood fuel consumption	HW_region	tons-wood/HH/year	0.90155	Calculation (Refer the f _{nrb} sheet)
Number of households consuming wood fuel	N_region	Number	43,812,055	Census Table (Refer the f _{nrb} sheet)
Commercial woody biomass consumption for energy applications	CE	tons-wood/year	0	Conservative assessment. Value not available
Commercial woody biomass consumption for non-energy applications	NE	tons-wood/yr	30,690,000	http://wipl.net.in/fsi_report_20011/final_2011.pdf#page=107
Annual Wood Consumption	H	tons	70,188,758	As per Equation (3)
Forest-land				
Extent of Forest Area	F_Forest	ha	6,282,600	India State of Forest Report 2019 (Table 1.2 Page No. 4).
Protected Area in Forest [1]	P_Forest	ha	3,880,000	India State of Forest Report 2019 (Table 1.2 Page No. 4)
MAI in Forest	MAI_Forest	m ³ /ha/yr	0.70	https://www.forests.tn.gov.in/app/webroot/img/Yield%20table%20for%20few%20tree%20species%20grown%20in%20farm%20settings.pdf
Average Density of Wood species in India [2]	NA	tons/m ³	0.79	https://en.wikipedia.org/wiki/List_of_Indian_timber_trees
MAI in Forest	MAI_Forest	tons/ha/yr	0.55	Calculated
RB Available from Forest	RB_Forest	tons/yr	1,322,087.5	As per Equation (4)
Wooded-land				
Extent of Other land	F_Other	ha	2,650,800	India State of Forest Report 2019 (Table 6.2 Page No. 108); Other land is termed as "Trees Outside Forest" in the published document
Protected Area under Other land	P_Other	ha	0	No data available, hence considered "NIL" as it will be conservative with respect to f _{NRB} calculation
MAI in Other land	MAI_Other	tons/ha/yr	0.55	MAI for other land is not available in published reports, hence it is considered same as that of forest. This is conservative as the trees coverage of forests would be much higher as compared to other land and hence higher MAI of forests.

RB Available from Other land	RB_Other	tons	1,458,665.48	As per Equation (4)
Total RB	RB	tons	2,780,753.04	Refer the fnrb sheet
Total NRB	NRB	tons	67,408,005.15	As per Equation (2)
f _{NRB}	NRB/(RB + NRB)	%	96.04%	As per Equation (1)

Sample Calculation for the 1st year of operation

$$B_{y,savings,i,j} \text{ (For the first year)} = 1.3^{17} \times (0.357/0.10-1)$$

$$= 3.344 \text{ Tones/Year}$$

$$ER_{y,i,j} \text{ (For the first year)}$$

$$= 3.344 \times 100,000 \times 0.9604 \times 0.0156 \times (112 + 26.23) \times 0.95$$

$$= 657,835 \text{ (Rounded down value)}$$

4.2 Project Emissions

The methodology does not account for project emissions and project emissions separately, but instead quantifies net emission reductions achieved by the project. Please refer to Section 4.4.

4.3 Leakage

Leakage is already accounted in calculation of emission reduction by adjusting with default value of 0.95.

4.4 Net GHG Emission Reductions and Removals

Year	Estimated baseline	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage	Estimated net GHG emission
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¹⁷ Assumption for ex-ante ER calculation, the parameter shall be monitored ex-post.

	emissions or removals (tCO ₂ e)		emissions (tCO ₂ e) ¹⁸	reductions or removals (tCO ₂ e)
Year 1 (01-10-2024 - 30-09-2025)	657,835	0	0	657,835
Year 2 (01-10-2025 - 30-09-2026)	1306,533	0	0	1306,533
Year 3 (01-10-2026 - 30-09-2027)	1,882,221	0	0	1,882,221
Year 4 (01-10-2027 - 30-09-2028)	2,393,088	0	0	2,393,088
Year 5 (01-10-2028 - 30-09-2029)	2,846,406	0	0	2,846,406
Year 6 (01-10-2029 - 30-09-2030)	2,590,799	0	0	2,590,799
Year 7 (01-10-2030 - 30-09-2031)	2,298,976	0	0	2,298,976
Year 8 (01-10-2031 - 30-09-2032)	1,723,288	0	0	1,723,288
Year 9 (01-10-2032 - 30-09-2033)	1,212,421	0	0	1,212,421

¹⁸ 5% leakage is already deducted and considered in the baseline calculation as per the methodology.

Year 10 (01-10- 2033 – 30- 09-2034)	7,59,103	0	0	7,59,103
Total	17,670,670	0	0	17,670,670

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	NCV _{woodfuel}
Data unit	TJ/tonne
Description	Net calorific value of the non-renewable woody biomass that is substituted or reduced
Source of data	Default as per applied approved methodology VMR0006, v1.1
Value applied	0.0156
Justification of choice of data or description of measurement methods and procedures applied	Default value as per applied approved methodology
Purpose of Data	Calculation of baseline emissions
Comments	The value is fixed ex-ante

Data / Parameter	EF _{wf,CO2}
Data unit	tCO ₂ /TJ
Description	CO ₂ emission factor for the use of wood fuel in baseline scenario
Source of data	Default as per applied approved methodology VMR0006, v1.1
Value applied	112
Justification of choice of data or description of measurement methods and procedures applied	Default as per applied approved methodology VMR0006, v1.1
Purpose of Data	Calculation of baseline emissions
Comments	The value is fixed ex-ante

Data / Parameter	EF _{wf,non CO2}
------------------	--------------------------

Data unit	tCO ₂ /TJ
Description	The non-CO ₂ emission factor for the use of wood fuel in baseline scenario
Source of data	Default as per applied approved methodology VMR0006, v1.1
Value applied	26.23
Justification of choice of data or description of measurement methods and procedures applied	Default as per applied approved methodology VMR0006, v1.1
Purpose of Data	Calculation of baseline emissions
Comments	The value is fixed ex-ante

Data / Parameter	$f_{NRB,y}$
Data unit	Fraction
Description	Fraction of non-renewable biomass saved by the project activity
Source of data	CDM Tool
Value applied:	0.9604
Justification of choice of data or description of measurement methods and procedures applied	This parameter shall be determined ex-ante and approved by the Government (Ministry of Environment). The calculation was undertaken by using the version 03 of Tool 30.
Purpose of the data	Calculation of baseline and project emissions
Comments	The report in terms of spread sheet of f_{NRB} will be made available to VVB during the validation.

Data / Parameter	η_P
Data unit	Fraction
Description	Efficiency of project stove at the start of project activity
Source of data	Manufacturer specification
Value applied	0.38
Justification of choice of data or description of measurement methods and procedures applied	The parameter has been sourced from the efficiency tested in laboratory. Testing results shall be submitted to VVB for validation.
Purpose of Data	Calculation of $\eta_{y,i,j}$
Comments	The first instance has Greenway Jumbo model and the efficiency is defined ex-ante. For any new model of ICS included in future instances, the efficiency shall be defined at the time of inclusion in group project.

5.2 Data and Parameters Monitored

Data / Parameter	$B_{y=1,new,i,j,survey}$
Data unit	tones/household/year
Description	Annual quantity of woody biomass that would have been used in the household in the absence of the project activity to generate useful thermal energy equivalent to that provided by the project devices
Source of data	Ex-post Surveys
Justification of choice of data or description of measurement methods and procedures applied	Minimum sample size of each type i and batch j should be in line with the latest version of Standard for sampling and surveys for CDM project activities and programme of activities or guidelines provided in methodology Section 8.4 option (b). Determined in the first year of the introduction of the devices (e.g. during the first year of the crediting period, $y=1$) through measurement campaigns at representative households and/or sample survey. Sample surveys to estimate this parameter, that are solely based on questionnaires or interviews (i. e. that do not implement measurement campaigns) may only be used if the following conditions are satisfied. (i) Baseline cookstoves have been completely decommissioned and only improved cookstoves are exclusively used in the project households; (ii) If multiple devices are used in the project, it is possible from the results of the survey questions to clearly differentiate the quantity of firewood being used by each device. In other words, if more than one device, or another device that consumes firewood, are in use in project households, then the sample survey needs to distinguish the quantity of firewood used by the project device and the other devices that use firewood.
Purpose of the data	Calculation of baseline
Comments	-

Data / Parameter	$N_{y,i,j}$
Data unit	Number
Description	Number of project devices of type I and batch j operating during year y
Source of data	Monitoring
Description of measurement methods and procedures to be applied	Measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve 95/10 confidence precision according to the latest version of Standard for sampling and surveys for CDM project activities and programme of activities.
Frequency of monitoring/recording	Annual
Value applied	For ex-ante emission reduction calculation, it is assumed that the project will distribute up to 500,000 ICS in phases i.e. 100,000 each year for first 5 years.
Monitoring equipment	Monitoring survey
QA/QC procedures to be applied	The project implementor shall maintain a distribution record to calculate this parameter.
Purpose of data	Calculation of baseline emissions
Calculation method	Proportion of operational stoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value
Comments	--

Data / Parameter	η_{old}
Data unit	Fraction
Description	Efficiency of baseline cookstove
Source of data	Methodological default value
Value applied	0.1
Justification of choice of data or description of measurement methods and procedures applied	A default value of 0.1 shall be used if baseline device is a three-stone fire using firewood (not charcoal), or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney
Purpose of Data	Calculation of baseline emissions
Comments	The value is fixed ex-ante

Data / Parameter	$\eta_{new,i,j}$				
Data unit	Fraction				
Description	Efficiency of the device of each type i and batch j implemented as part of the project activity				
Source of data	Manufacturer specification or <i>Certification by a national standards body or an appropriate certifying agent recognized by that body</i>				
Description of measurement methods and procedures to be applied	To adopt Option V given in the methodology: “Efficiency of the improved cookstoves to be estimated using equation 5 of applied approved methodology where loss in efficiency per year is calculated, and therefore this parameter does not need to be monitored”				
Frequency of monitoring/recording	Annual				
Value applied	0.38 ¹⁹ at start of operation <table border="1"> <thead> <tr> <th>Year</th><th>Stove efficiency</th></tr> </thead> <tbody> <tr> <td>1</td><td>0.357</td></tr> </tbody> </table>	Year	Stove efficiency	1	0.357
Year	Stove efficiency				
1	0.357				

¹⁹ The efficiency indicated is for Jumbo models to be included in first project of the grouped project, any new model included in the grouped projects shall demonstrate the efficiency at the time of inclusion following the requirements of VMR0006, v 1.1

	2	0.354
	3	0.350
	4	0.347
	5	0.343
	6	0.340
	7	0.336
Monitoring equipment	Not applicable	
QA/QC procedures to be applied	Not applicable	
Purpose of data	Calculation of baseline emissions	
Calculation method	As per equation 5 of VMR0006, version 1.1 i.e. $\eta_{new,i,j} = \eta_P \times (DF_n)^{y-1} \times 0.94$	
Comments	--	

Data / Parameter	Life span
Data unit	Years
Description	The operating lifetime of the project device. The life span should be reported if the methodology equation 5 is adopted to determine the project stove efficiency
Source of data	Manufacturer specification
Description of measurement methods and procedures to be applied	The data source as per manufacturer specification
Frequency of monitoring/recording	Once at the time of installation of batch of type of stove
Value applied	7
Monitoring equipment	Not applicable

QA/QC procedures to be applied	Not applicable
Purpose of data	Calculation of baseline emission
Calculation method	Not applicable
Comments	--

Data / Parameter	Date of commissioning of batch j
Data unit	Date
Description	To establish the date of commissioning, the Project Participant may opt to group the devices in “batches” and the latest date of commissioning of a device within the batch shall be used as the date of commissioning for the entire batch.
Source of data	Project database
Description of measurement methods and procedures to be applied	Fixed and recorded at the time of commissioning/distribution of the last project device in the batch
Frequency of monitoring/recording	Fixed and recorded at the time of commissioning/distribution of the last project device in the batch
Value applied	01/06/2023 (expected)
Monitoring equipment	Not applicable
QA/QC procedures to be applied	Not applicable
Purpose of data	Calculation of emission reduction
Calculation method	Not applicable
Comments	--

5.3 Monitoring Plan

The methods for measuring, recording, storing, aggregating, collating and reporting data and parameters. Where relevant, include the procedures for calibrating monitoring equipment.

The organizational structure, responsibilities and competencies of the personnel that will be carrying out monitoring activities.

The policies for oversight and accountability of monitoring activities.

The procedures for internal auditing and QA/QC.

The procedures for handling non-conformances with the validated monitoring plan.

Any sampling approaches used, including target precision levels, sample sizes, sample site locations, stratification, frequency of measurement and QA/QC procedures.

Sample Plan:

The monitoring plan is designed to monitor the parameters listed in Section above, which are required for calculation of the actual GHG emission reduction achieved by the project activity using ex post sampling survey. The share of operating stoves and the continued use of pre-project devices will be determined based on sampling procedures as outlined below. The PP will be responsible for conducting the sampling surveys and maintaining a database with all operating stoves.

No monitoring for leakage through competitive uses of biomass is required, as the parameter wood usage is multiplied by 95% to account for leakage.

As per the Guideline for Sampling and Surveys for CDM Project Activities and Programme of Activities, version 04, the sampling plan is the following:

(a) Sampling Design

Due to the large number of ICS envisioned to be distributed as part of the project activity, it is not economically feasible to monitor each individual ICS unit distributed. Therefore, representative sampling will be undertaken as part of a project instance-wide Sampling Plan (by grouping and sampling across project activities) that is designed in line with the requirements of the “Sampling and surveys for CDM project activities and programme of activities”, version 04.

(i) Objective and Reliability Requirements:

The objective is to obtain an unbiased and reliable estimate of the proportion or mean value of the following key variables over the course of the crediting period. As per CDM Methodology AMS-II.G, 90/10 confidence/precision shall be applied for annual sampling requirement and 95/10 for biennial sampling inspection. As per Standard for Sampling and Surveys for CDM Project

Activities and Programme of Activities, 90/10 confidence/precision to be adopted for small-scale project and 95/10 for large-scale project. Given that the size of the project is under the category of large projects, hence 95/10 confidence/precision shall be adopted for all parameters unless the average annual emission reductions of the project below the threshold:

Monitored Parameter:

Parameter	Description	Parameter type/Frequency
$B_{y=1,new,i,j,survey}$	Quantity of woody biomass used by project devices in tonnes per device of type i. and batch j	Once Before the First Verification
$N_{y,i,j}$	Number of project devices of type i and batch j operating during year y	Proportion/Annual

(ii) Target Populations:

- The target population for the proportion of ICS still in operation $N_{y,i,j}$ and $B_{y=1,new,i,j,survey}$ (during first verification only) of this project are all households in the project database which are using fuel wood in ICS distributed under the project for cooking.

(iii) Sampling Frame

To ensure the homogeneity of the project instances included for a single sampling plan, two sampling frames shall be defined. Overall, all project instances will have same group of end users which is from households from rural/semi-urban areas. The projects are to be implemented in rural/semi-urban areas; thus, it is expected that the geographical locations do not have influence on the parameter of interest. Therefore, all above mentioned parameters can be assumed to be highly homogeneous for each ICS model regardless of how the end user group and distribution/installation location is defined.

1) Sampling frame for proportion of ICS in operation ($N_{y,i,j}$)

The sample frame refers to all the information sources on the Database. There are two primary mechanisms for data collection: the Registration Process for newly distributed/installed ICS and the Monitoring Survey (which includes a household questionnaire and visual inspection of ICSs) that will be used throughout the lifetime of the ICS. The detailed information collected from Registration Process is used to populate the stoves Database and the Monitoring Survey follows “Sampling and Surveys for CDM Project Activities and Programme of Activities”, version 04.

As explained below (on section “Sampling Method”), to take the different characteristics of different project instances Implemented and ICS models into consideration, Project instances

shall be grouped together to create a Primary Sampling Unit, which is homogenous. As per EB 86 Annex 04, Appendix-2, paragraph 1, for the use of a single sampling plan covering a group of projects, provided the homogeneity of population can be demonstrated, or differences are considered in the sample size calculation, a 95/10 confidence/precision is applied for biennial sampling.

The first step is to identify the Primary Sampling Units. Primary sampling units are project instances, which have:

1. The same ICS models

That is project with the same ICS model can therefore be grouped together and form a Primary Sampling Unit. In case project has two different ICS models being implemented – this will form two Primary Sampling Units. This is justified by the fact that project might vary in terms of performance and it is important for the PP to collect and monitor accurate data for each stove model.

In line with applied approved methodology AMS-II.G, as installing data logger is not practical and if any use of pre project device can be monitored in a common survey with other monitoring parameters; therefore, a random sub-sample within the common survey can be taken to determine continued use of old cookstoves and its proportional usage by including suitable questionnaire.

As mentioned above the either separate sampling can be done for this parameter or can be clubbed with sampling of above parameters, wherein interview question to be included to determine the average fuel wood used by the project device.

(iv) Sampling Method

The sampling method for monitored parameters $N_{y,i,j}$, and $B_{y=1,new,i,j,survey}$ is Simple Random Sampling and samples will be randomly selected from the primary sampling units as illustrated above. To ensure a random selection of ICS, random number generators shall be applied. Each ICS in the target population is uniquely identifiable by its unique ID number. Each ICS can thus be allocated a Sample Selection Number in each monitoring period, starting at 1 and increasing up to the total number of ICS in the Database for that pre-defined sampling frame. Applying the random number generators, the ICS can then be randomly chosen from the defined population up to the required sample size as calculated by the PP.

To determine the parameters, sampling will involve the following approaches (outcome in brackets):

$N_{y,i,j}$: Visual inspection of the premises to see if ICS is operational and in use. Interview with end user if required to verify that ICS is still in use (Yes/No)

$B_{y=1,new,i,j,survey}$: Quantity of woody biomass used by project devices in tonnes per device (first year of installation)

Using the formulas in the section “Sample Size” below, the PP will randomly sample the required number of ICS from the primary sampling units.

(v) Sample Size

For the estimation of the proportion or mean value of the parameters investigated, the minimum sample size for each sample frame has to achieve the 95/10 confidence/precision. for annual and 95/5 confidence/precision for biennial sampling.

The procedure to determine the sample of households will ensure that they adequately represent the broader project population, minimizing sampling error. Using, a 95 per cent confidence level, and a 5 per cent margin of error, random samples will be selected from each Primary Sampling Unit. There are three parameters that will be estimated through sampling: the number of stoves still in operation during the monitoring period as determined by the monitoring survey ($N_{y,i,j}$) and Quantity of woody biomass used by project devices in tonnes per device ($B_{y=1,new,i,j,survey}$),. In line with sampling guidelines, all can be sampled in a single survey with a random sample of households using the above-described confidence/precision levels depending on annual or biennial monitoring frequency. The $N_{y,i,j}$ requires proportion/percentage parameters, however the wood consumption by ICS may have variation and will be a mean value.

In order to calculate the required sample size estimates, values for the proportions, mean values, and standard deviations are required. As per Guidelines for Sampling and surveys for CDM project activities and programmes of activities, version 04.0 Appendix 1 paragraph 5, there are different ways available to obtain the estimates of the parameter of interest:

- (a) Refer to the result of previous studies and use these results;
- (b) In a situation where information from previous studies is not available, a preliminary sample as a pilot could be conducted and use that sample is used to provide the estimates;
- (c) Use best guesses based on the researcher's own experiences.

For the registration purpose of project, option C shall be applied. For the first monitoring period, values from a pilot shall be applied. For the following monitoring periods, the estimates shall be adjusted taken into account the results of the previous monitoring period(s) or the result from recent pilot study, which is conducted after the previous monitoring periods.

To estimate the sample size for parameters $N_{y,i,j}$ & $B_{y=1,new,i,j,survey}$ the following equation is used:

$$n \geq \frac{1.96^2 N \times p (1-p)}{(N - 1) \times 0.1^2 \times p^2 + 1.96^2 \times p(1-p)}$$

Where:

n = Sample size

N = Population size (Total number of households/ICS)

p = Expected proportion

1.96 = Represents the 95% confidence required

(In the case of 90% confidence, 1.645 shall be used)

0.1 = Represents the 10% relative precision

The following assumptions are made to exemplify the sample size calculation for parameters:

$N_{y,i,j}$ and $B_{y=1,new,t,j,survey}$

The PP envisage distribution of 500,000 ICSs over next five year. Hence, population size, N , is taken as 500,000 households/ICS (Assuming one ICS for one household).

It is expected at least 90% of ICS will be operational, hence the expected proportion p for $N_{y,i,j}$ is taken as 0.9.

Sample size calculation:

The calculation of the required sample size for each parameter in the first monitoring period is illustrated below for a 95/10 level of confidence and precision (for biennial monitoring periods the sample sizes will be recalculated using 95/5 confidence/precision values). In all cases a conservative approach is taken, however if for any parameter the required 95/10 confidence/precision is not met then the project proponent will randomly select an additional sample and collect further data from this sample to ensure the pooled data meet or exceed the required thresholds.

Parameter $N_{y,i,j}$: Based on the above assumptions, the resulting sampling size for a 95/10 confidence/precision is calculated as:

$$n \geq \frac{1.96^2 \times 500,000 \times 0.9 (1-0.9)}{(500,000-1) \times 0.1^2 \times 0.9^2 + 1.96^2 \times 0.9 (1-0.9)}$$

Which comes out to be

$$n \geq 42.68$$

$$n = 43$$

Therefore, in this case a sample size of 43 is to be sampled from each primary sampling unit for annual sampling.

This above demonstrated calculation is an example for annual sampling, however the actual sample size will be calculated prior to monitoring survey with actual population and actual monitoring duration i.e. annual or biennial.

In case the resulting sample size to achieve the desired confidence/precision levels is smaller than 30 ICS, a minimum sample size of 30 shall be chosen when the parameter of interest is a proportion in line with para 14 of Standard: Sampling and surveys for CDM project activities and programmes of activities, Version-09.